

ECE 592-091 - Resonant Power Converters

Lab 4: Prototyping and Open-Loop Testing of a DC-DC Series-Resonant Converter (SRC)

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Objective

The objective of this lab is to teach students how to:

- Measure relevant characteristics of the converter's transformer.
- Assemble a series-resonant converter and experimentally determine its DC-DC voltage gain function MV and efficiency η
- Combine measurements and calculations to determine loss distribution among different converter components.

Task 1: Characterising the Transformer and Re-Tuning the Inverter

Aim

Measure the magnetizing inductance of the transformer at 100 kHz, referred to as the transformer's primary. Calculate the mutual inductance between the primary and each secondary winding. Explain the procedure and present the measured and calculated values in your report.

Procedures and Results

1. Using the Adjustment (ADJ) tab of the LCR meter, conduct short- and open-circuit tests to determine corrections the instrument will use to compensate for the measurement error. Before conducting the tests, set the meter's cable length to 1 m.
2. Using the procedure explained in the LCR application guide (pages 17-18), measure the magnetizing inductance of the transformer at 100 kHz, referred to the primary of the transformer. Calculate the mutual inductance between the primary and each secondary winding. Explain the procedure and present the measured and calculated values in your report.

Figure 1.1 illustrates a transformer electrical equivalent circuit model, including series inductors L_{l1} and L_{l2} which model the leakage inductances. These leakage inductances cause the terminal voltage ratio $V_2(t)/V_1(t)$ to differ from the ideal turns ratio n_2/n_1.

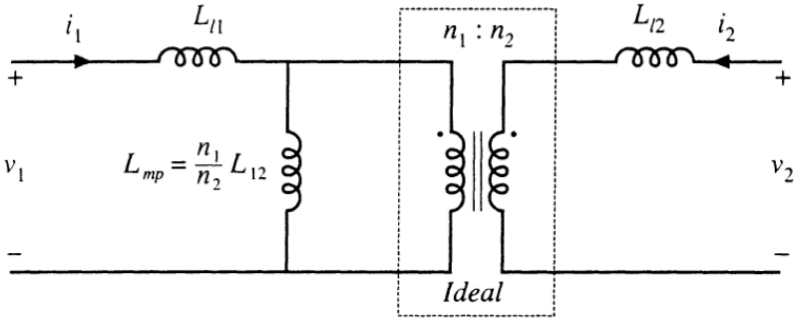


Figure 1.1: Two-winding transformer equivalent circuit, including magnetizing inductance referred to primary, and primary and secondary leakage inductances

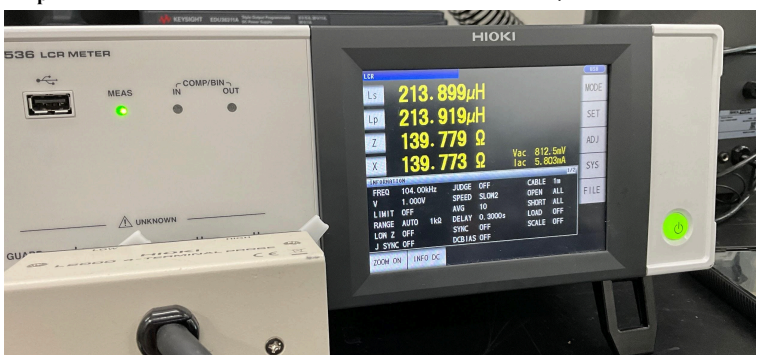
The terminal equations of a two-winding transformer is the following:

$$\begin{bmatrix} v_1(t) \\ v_2(t) \end{bmatrix} = \begin{bmatrix} L_{11} & L_{12} \\ L_{12} & L_{22} \end{bmatrix} \frac{d}{dt} \begin{bmatrix} i_1(t) \\ i_2(t) \end{bmatrix} \qquad (12.48)$$

L_{12} is the mutual inductance, and is given by:

$L_{12} = \frac{n_1 n_2}{R} = \frac{n_2}{n_1} L_{mp}$; where L_{mp} is the magnetizing inductance

L_{mp} was measured at 104kHz to be **213.899μH**



Hence, the **mutual inductance** $L_{12} = \frac{1}{1.2} * 213.899 * 10^{-6} = 178.2492\ \mu H$

Calculating mutual inductance using measurement values:

La	Lo	Mutual Inductance
778μH	7.8μH	192.55μH
780μH	7.82μH	193.045μH

Noted that values were borrowed from another group since the group missed proper measurement of La and Lo according to the LCR application guidelines.

3. Using the procedure presented in the LCR application guide (page 17), calculate the equivalent leakage inductance and the equivalent resistance of the transformer at 100 kHz, referred to the primary side. Calculate the two parameters independently for each secondary winding. Explain the procedure and present the measured and calculated values in your report.

To obtain primary winding measurement, both secondary windings are shorted; to measure secondary A winding, primary and secondary B are shorted; to measure secondary B winding, primary and secondary A windings are shorted. The following table are the results from the procedures:

Measurement windings	Ls (uH)	Rs (mOhm)
Primary	0.58189	35.82
Secondary A	0.8823	42.34
Secondary B	0.8430	60.86


HIOKI
0.58196μH
0.58713μH
35.84mΩ
4.07125 Ω
Vac 5.086mV
Iac 9.888mA
LOCAL


IM3536 LCR METER
HIOKI
0.88230μH
0.88706μH
42.34mΩ
7.89220 Ω
Vac 7.020mV
Iac 8.891mA
LOCAL


HIOKI
0.84301μH
0.85330μH
60.86mΩ
5.04655 Ω
Vac 6.773mV
Iac 9.867mA
LOCAL

4. Using the resistive procedures presented in the LCR application guide (page 19), calculate the transformer turn ratio at 100 kHz for each secondary. Use the resistive load bank to create 5Ω resistance. To improve the accuracy, measure the exact value of the resistance before testing. Explain the procedure and present the measured and calculated values in your report.

Noted that the resistance bank was changed to **5Ω** in the experiment on the secondary side of the transformer. To find turn ratio, N, the following equations were used:

$$V_2 = NV_1$$

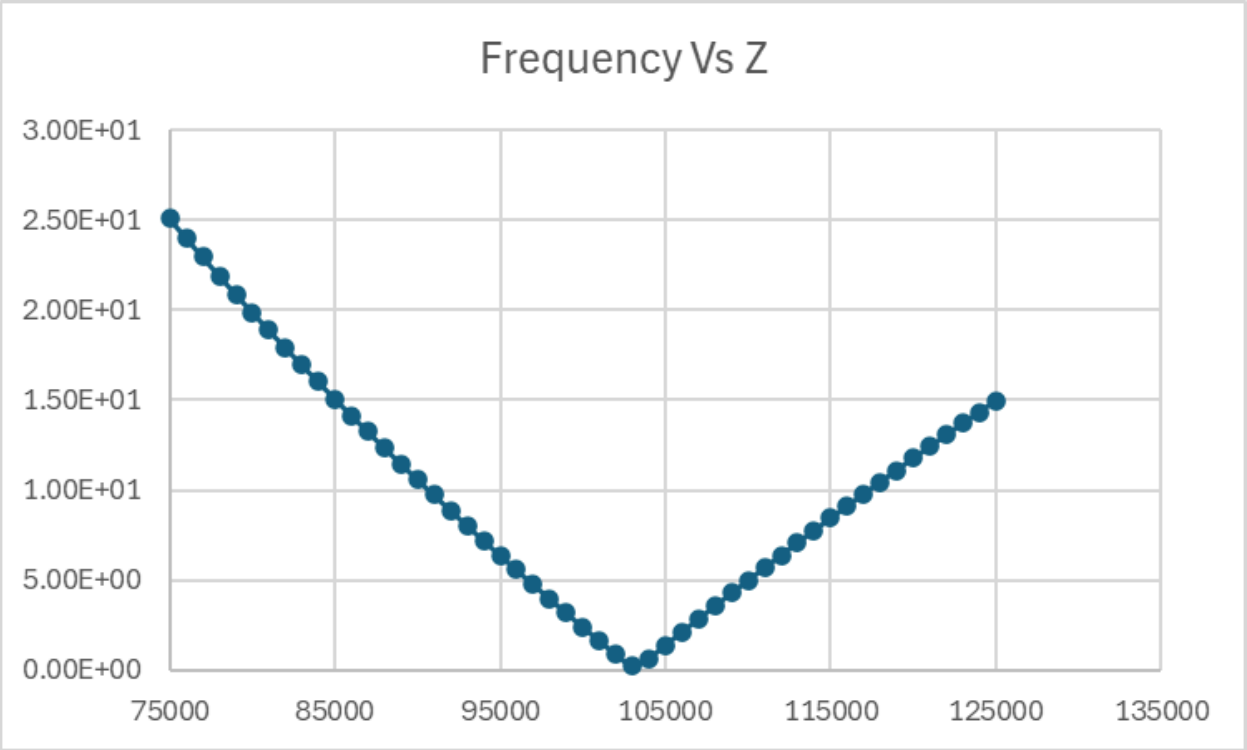
$$I_2 = \frac{I_1}{N}$$

$$Z_2 = \frac{V_2}{I_2} = NV_1\left(\frac{N}{I_1}\right) = N^2 Z_1$$

$$N = \sqrt{\frac{Z_2}{Z_1}} = \sqrt{\frac{7.237}{5}} = 1.203 = \frac{N_p}{N_s}$$

The turn ratio is **1.2:1**

5. Solder the transformer to the ITR board. Shorten one of the secondary windings and use the LCR meter to sweep the input inductance and resistance of the resonant network in the range from 75 kHz to 125 kHz. How much do the transformer parasitics change the resonant frequency of the resonant network? How much do the losses in the transformer contribute to the losses in the resonant network? If necessary, consider retuning the network by adding, removing, or substituting the parallel capacitors Cr. Include in your report the measured curves followed by requested comments.



Resonant frequency = 103.2kHz

- How much do the transformer parasitics change the resonant frequency of the resonant network?

  The initial resonant frequency of the circuit is 104KHz. When the transformer is added in, the resonant frequency is reduced to 103.2KHz.The reason behind this decrease is the parasitic capacitance and inductance that are introduced because of the transformer. Parasitic capacitance is due to the interwinding gaps. Parasitic inductance can be observed because of the leakage inductance in the transformer. The leakage inductance acts as another series inductance, which in total increases the ineffective inductance value which ultimately decreases the frequency.

  For parasitic capacitance, the capacitance between two windings adds up to the effective capacitance, which reduces the frequency.

- How much do the losses in the transformer contribute to the losses in the resonant network?

  The losses in the transformer can reduce the quality factor, the heat generated within the transformer can impact the operating temperature of other components in the resonant network, potentially leading to performance degradation or even damage. Transformer losses can reduce overall efficiency due to heat dissipation.

Task 2: Converter assembling and measuring the DC-DC voltage gain and efficiency

Aim

To utilize primary and auxiliary power supplies, make appropriate connections, assemble the ITR board, and probe the input and output currents and voltages to find the voltage transfer function Mvr and efficiency η.

Procedures

1. Resolder the connections of the resistive bank to match the RL value for the planned experiment from Table II.

2. Use the multimeter to measure the resistance of the resistive assembly and store the measured value in Table II.

3. Connect the main and auxiliary power sources (both in OFF mode), as shown in Fig. 4.

4. Connect the microcontroller to the computer using the USB cable.

5. Connect the multimeter to measure the load voltage and current.

6. Verify the circuit connections with the TA.

7. Open **CCS** and the project file.

8. Compile the **CCS** code and reprogram the microcontroller.

9. Add the frequency variable **Fref_kHz_input** into the Expression section of the CCS Debug window.

10. Activate the Continuous Refresh to track the actual value of variables in the Debug window.

11. Set the frequency to 75 kHz.

12. Set the auxiliary power supply voltage to **12 V** and its output current limit to **0.2 A**.

13. Turn the auxiliary power supply ON.

14. Press the **Reset** protection button on the PCB.

    a. This action resets the initial fault caused by the startup transient.

    b. LED DL7 will light up after the reset.

15. Cover the power board with the safety enclosure.

16. Wear **safety glasses**.

17. Set the main power supply voltage to 0 V and its output current limit to 2 A.

18. Turn the main power supply ON.

19. Slowly increase the main power supply voltage until it reaches the input voltage specified in Table II for the particular experiment.

20. Change the frequency in the following steps:

    a. 5-kHz steps from 75 kHz to 95 kHz

    b. 1-kHz steps from 95 kHz to 105 kHz

    c. 5-kHz steps from 105 kHz to 125 kHz

21. At each frequency step:

    a. Record the output voltage and power from the oscilloscope.

    b. Record the input power from the main power supply.

    c. Use the measured values to calculate efficiency and DC-DC voltage gain.

22. Change the input voltage and/or load resistance according to the order in Table II, and repeat the above step.

23. Include in your report:

a. Table II with all values filled in.

b. Plots showing MV = f(frequency) and η = f(frequency) for three experiments defined in Table II.

24. We used a voltmeter to measure the voltage and already knew load R and were able to find the current.

Results

Case 1: Vin=36V and RL=8.33 Ω

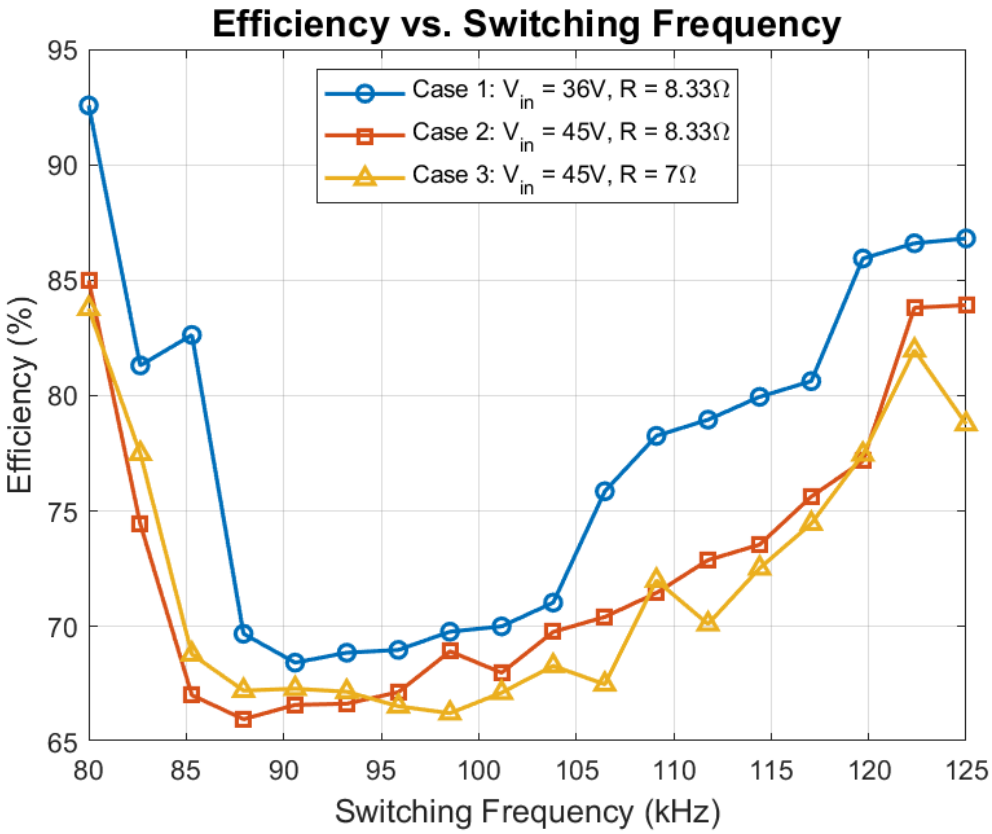
Frequency(Hz)	V_supply	I_supply	Vout	I_out	Mvf (Vout/Vin)	Pout	Pin	Efficiency	Efficiency(%)
75000	36	NA							
80000	36	0.303	8.78	1.15	0.2439	10.10	10.91	0.9257	92.57
85000	36	0.438	10.34	1.24	0.2872	12.82	15.77	0.8131	81.31
90000	36	0.559	10.73	1.28	0.2981	13.73	20.12	0.6825	68.25
95000	36	0.59	11.13	1.33	0.3092	14.80	21.24	0.6969	69.69
96000	36	0.584	10.9	1.32	0.3028	14.39	21.02	0.6844	68.44
97000	36	0.561	10.75	1.29	0.2986	13.87	20.20	0.6866	68.66
98000	36	0.542	10.6	1.27	0.2944	13.46	19.51	0.6899	68.99
99000	36	0.52	10.45	1.25	0.2903	13.06	18.72	0.6978	69.78
100000	36	0.503	10.41	1.235	0.2892	12.86	18.11	0.7100	71.00
101000	36	0.482	10.36	1.225	0.2878	12.69	17.35	0.7314	73.14
102000	36	0.461	10.32	1.22	0.2867	12.59	16.60	0.7586	75.86
103000	36	0.442	10.18	1.2	0.2828	12.22	15.91	0.7677	76.77
104000	36	0.423	10.02	1.2	0.2783	12.02	15.23	0.7896	78.96
105000	36	0.403	9.83	1.18	0.2731	11.60	14.51	0.7995	79.95
110000	36	0.32	8.93	1.07	0.2481	9.56	11.52	0.8294	82.94
115000	36	0.257	8.11	0.97	0.2253	7.87	9.25	0.8503	85.03
120000	36	0.204	7.26	0.87	0.2017	6.32	7.34	0.8600	86.00
125000	36	0.155	6.34	0.764	0.1761	4.84	5.58	0.8681	86.81

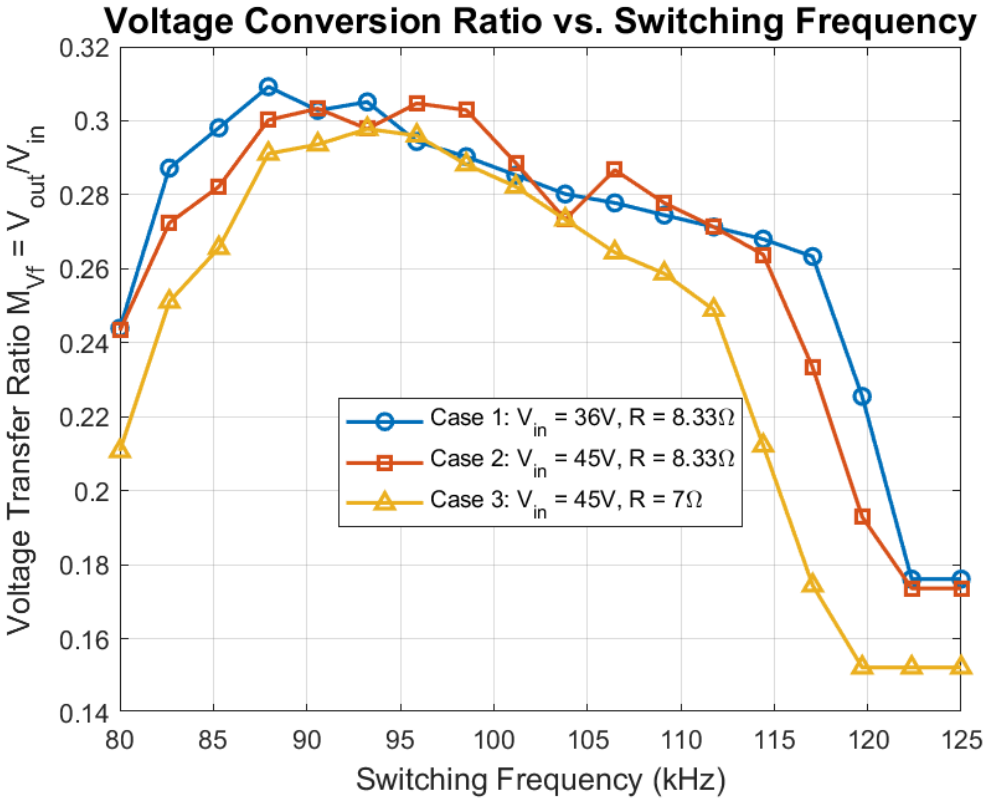
Case 2: Vin=45V and RL=8.33 Ω

Frequency(Hz)	V_supply	I_supply	Vout	I_out	Mvf (Vout/Vin)	Pout	Pin	Efficiency	Efficiency(%)
75000	45	NA							
80000	45	0.375	10.95	1.31	0.2433	14.34	16.88	0.8500	85.00
85000	45	0.538	12.26	1.47	0.2724	18.02	24.21	0.7444	74.44
90000	45	0.64	12.7	1.52	0.2822	19.30	28.80	0.6703	67.03
95000	45	0.737	13.51	1.62	0.3002	21.89	33.17	0.6599	65.99
96000	45	0.748	13.65	1.64	0.3033	22.39	33.66	0.6651	66.51
97000	45	0.749	13.7	1.64	0.3044	22.47	33.71	0.6666	66.66
98000	45	0.744	13.71	1.64	0.3047	22.48	33.48	0.6716	67.16
99000	45	0.73	13.63	1.64	0.3029	22.35	32.85	0.6805	68.05
100000	45	0.703	13.46	1.62	0.2991	21.81	31.64	0.6893	68.93
101000	45	0.669	13.21	1.59	0.2936	21.00	30.11	0.6977	69.77
102000	45	0.631	12.9	1.55	0.2867	20.00	28.40	0.7042	70.42
103000	45	0.587	12.5	1.51	0.2778	18.88	26.42	0.7146	71.46
104000	45	0.551	12.21	1.48	0.2713	18.07	24.80	0.7288	72.88
105000	45	0.512	11.85	1.43	0.2633	16.95	23.04	0.7355	73.55
11000	45	0.381	10.5	1.26	0.2333	13.23	17.15	0.7717	77.17
115000	45	0.304	9.97	1.15	0.2216	11.47	13.68	0.8381	83.81
120000	45	0.243	8.68	1.04	0.1929	9.03	10.94	0.8255	82.55
125000	45	0.194	7.81	0.938	0.1736	7.33	8.73	0.8392	83.92

Case 3: Vin=45V and RL=7 Ω

Frequency(Hz)	V_supply	I_supply	Vout	I_out	Mvf (Vout/Vin)	Pout	Pin	Efficiency	Efficiency(%)
75000	NA								
80000	45	0.342	9.48	1.36	0.2107	12.89	15.39	0.8377	83.77
85000	45	0.525	11.3	1.62	0.2511	18.31	23.63	0.7749	77.49
90000	45	0.66	11.95	1.71	0.2656	20.43	29.70	0.6880	68.80
95000	45	0.814	13.1	1.88	0.2911	24.63	36.63	0.6723	67.23
96000	45	0.838	13.29	1.91	0.2953	25.38	37.71	0.6731	67.31
97000	45	0.851	13.4	1.92	0.2978	25.73	38.30	0.6718	67.18
98000	45	0.854	13.32	1.92	0.2960	25.57	38.43	0.6655	66.55
99000	45	0.828	13.1	1.88	0.2911	24.63	37.26	0.6610	66.10
100000	45	0.808	12.95	1.86	0.2878	24.09	36.36	0.6625	66.25
101000	45	0.765	12.7	1.82	0.2822	23.11	34.43	0.6714	67.14
102000	45	0.72	12.3	1.8	0.2733	22.14	32.40	0.6833	68.33
103000	45	0.666	11.9	1.7	0.2644	20.23	29.97	0.6750	67.50
104000	45	0.6	11.64	1.67	0.2587	19.44	27.00	0.7200	72.00
105000	45	0.568	11.2	1.6	0.2489	17.92	25.56	0.7011	70.11
110000	45	0.395	9.55	1.35	0.2122	12.89	17.78	0.7253	72.53
115000	45	0.307	8.63	1.24	0.1918	10.70	13.82	0.7746	77.46
120000	45	0.241	7.85	1.13	0.1744	8.87	10.85	0.8179	81.79
125000	45	0.189	6.85	0.978	0.1522	6.70	8.51	0.7877	78.77


Efficiency vs. Switching Frequency
Case 1: V_in = 36V, R = 8.33Ω
Case 2: V_in = 45V, R = 8.33Ω
Case 3: V_in = 45V, R = 7Ω
Efficiency (%)
Switching Frequency (kHz)
95
90
85
80
75
70
65
80
85
90
95
100
105
110
115
120
125


Voltage Conversion Ratio vs. Switching Frequency
Case 1: V_in = 36V, R = 8.33Ω
Case 2: V_in = 45V, R = 8.33Ω
Case 3: V_in = 45V, R = 7Ω
Voltage Transfer Ratio M_Vf = V_out/V_in
Switching Frequency (kHz)
0.32
0.3
0.28
0.26
0.24
0.22
0.2
0.18
0.16
0.14
80
85
90
95
100
105
110
115
120
125

Experiment	Vin (V)	RL (targeted) Ohm	RL (measured) Ohm
1	36	8.33	8.373
2	45	8.33	8.373
3	45	7	7.228

Observations and Remarks

In this experiment, a critical aspect of loss measurement was overlooked. The wire connecting the board and resonant tank to the load introduced intrinsic losses, further alleviated by a loose connection. Ideally, losses should have been measured with the connector wires attached directly to the load bank using the LCR meter, which would have provided a clearer representation of the actual loss profile. As a result of this oversight, a degradation and/or an erratic pattern were observed in the output power (Pout), and consequently, in the voltage transfer ratio (Mvr) and efficiency. The same approach was tried with the current readings using the V^2/R_Load approach for output power and subsequent calculations were done as well. However, the results did not vary by a significant margin, thus validating that there was an issue during the hardware testing, which caused the errors in the readings.

Task 3: Measuring the loss distribution

Aim

In the task below, the input power PIN, the inverter current iinv(t), the DC output voltage and current, VO and IO, and the rectifier diode current was measured to reconstruct the losses in the converter

Procedures

1. The same setup as task 2 is used. We add a second current probe to the secondary side to measure the diode current in the oscilloscope.
2. The RMS value of iinv(t) current and the RMS and average values of the diode current were to be calculated.
3. Frequency was changed in the CCS debug window from a range of 98 to 105 Khz to study the losses between non-ZVS and ZVS switching
4. The measured values and necessary datasheets were used to calculate the total loss of the converter and the individual losses of resonant inductors and capacitors, transformer, diode rectifiers, filter capacitors, and MOSFET conduction losses.

Results

Frequency	Vout	Iout	Pin	Pout	Efficiency	Iinv	Total losses
98	13.32	1.92	38.43	25.5744	0.6654800937	1.6	12.8556
100	12.95	1.85	36.36	23.9575	0.6588971397	1.541666667	12.4025
102	12.3	1.8	32.4	22.14	0.6833333333	1.5	10.26
105	11.2	1.6	27	17.92	0.6637037037	1.333333333	9.08

NOTE: These values are off from the original values since there was a loose connection to the Load wires of the resistive bank causing the efficiency to be low even when ZVS was achieved at 102 KHZ.

The Values of the Components that contribute to the losses are given below:

Diode losses	rL	rCr(resonant capacitor)	rCf(Filter capacitor)	rDSon(MOSFET)	Transformer (secondary)
Rf=0.0175Ω	0.04Ω	0.015Ω	3.8Ω	0.007Ω	0.1032Ω
Vf=0.53V					
If=30A					

These Values were taken from the Analysis during the previous Lab. For the transformer the total secondary resistance,which was found out in Task1 of Lab 4.

Condition 1:
At 98 KHz, i inv(t),

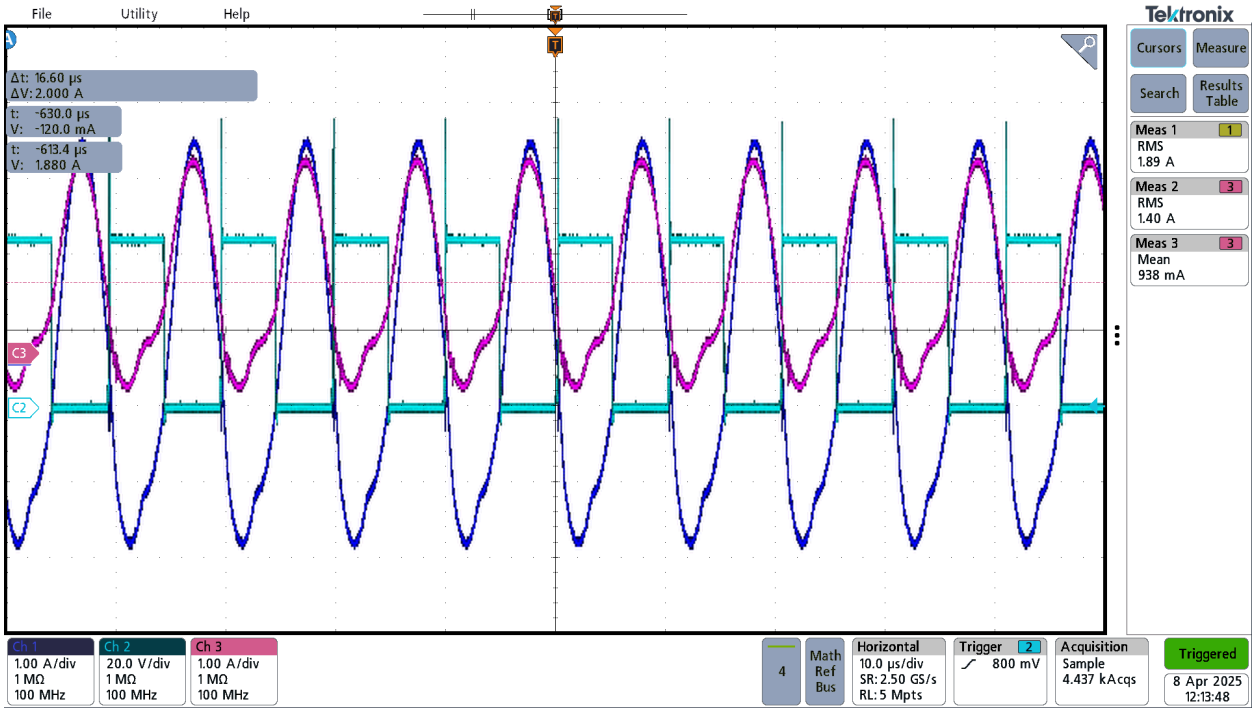


Figure: Waveform of inverter current, diode current and voltage across the switch.

Channel 1 represents the inverter current.
Channel 2 represents the voltage across the switch.
Channel 3 represents the current across the switch.

Measured RMS value of the inverter current is 1.89 A
Measured RMS value of the diode current is 1.40 A
Measured average value of the diode current is 938 mA

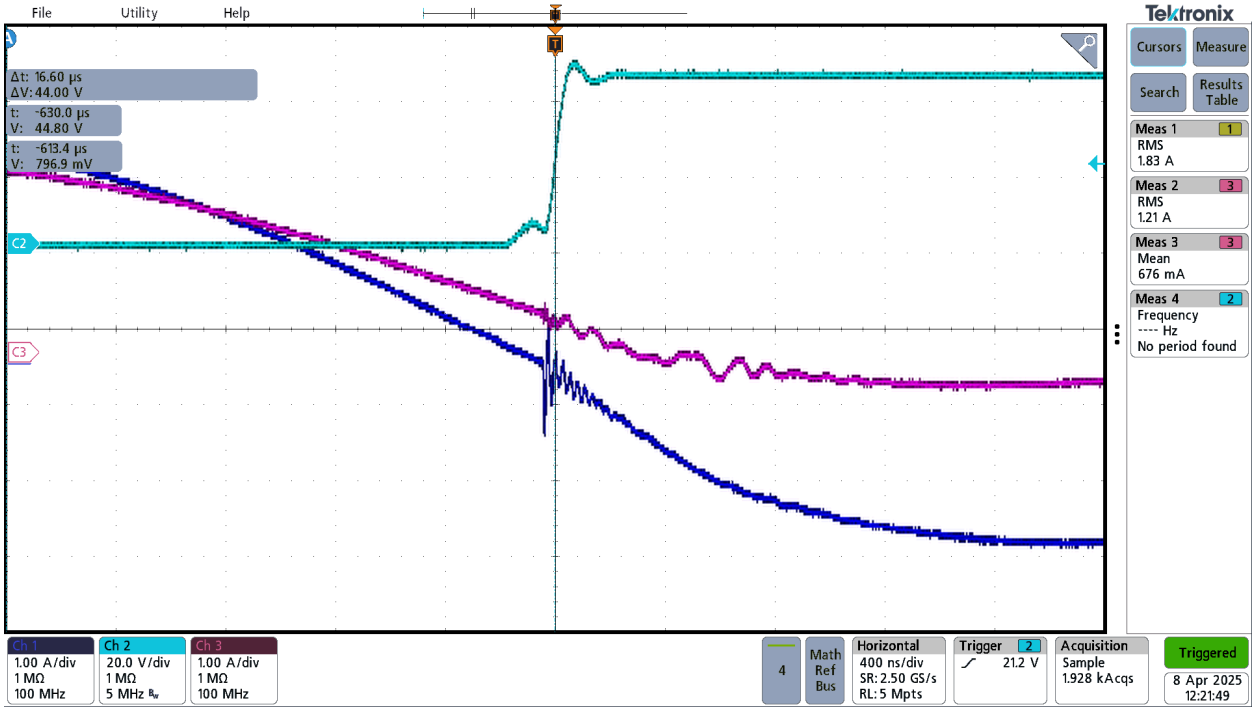


Figure: Hard switching Waveform at 98 Khz.

Measured RMS value of the inverter current is 1.83 A
Measured RMS value of the diode current is 1.21 A
Measured average value of the diode current is 675 mA

LOSS CALCULATION:

Total losses= 12.8586 W

The formula for losses are:

Resonant Inductor:

$$P_l = r_l \cdot \frac{I^2_{inv}}{2}$$

$$I_{inv} = I_{out} / n$$

$$where,$$

$$n = 1.2$$

Resonant Capacitor & Filter Capacitor:

$$P_{cr} = r_{cr} * I_{inv}{}^2 * 0.5$$

$$P_{rcf} = r_{cf} * I_o{}^2 (\pi^2/8 - 1)$$

MOSFET Conduction Losses:

$$P_{mosfet} = 2 * (I_{inv}{}^2 * r_{ds(on)}) / 4$$

Diode Losses:

$$P_{diode} = 2\left(r_f \cdot I_{drms}{}^2 + V_f \cdot I_o/2 \right)$$

Transformer Losses:

$$P_t = I_{drms}{}^2 \cdot R_{sec}$$

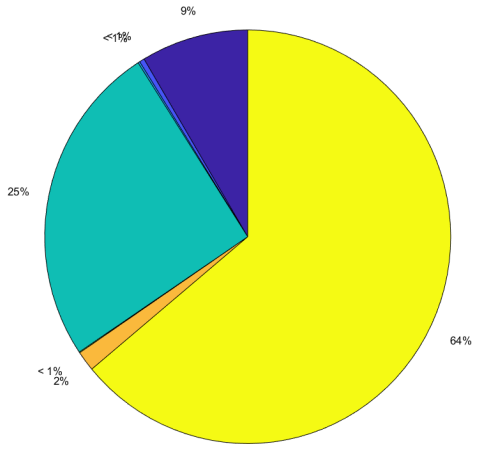
MOSFET Switching losses:

Switching loss=Total power loss- Losses of remaining components

Using these Formula the Losses for 98 Khz was found out to be:

Frequency (HZ)	**Diode Loss (W)**	**Inductor Loss (W)**	**Resonant capacitor Loss(W)**	**Filter Capacitor Loss(W)**	**MOSFET Conduction Loss(W)**	**Transformer Losses(W)**	**MOSFET Switching Losses(W)**
98	1.0971	0.0512	0.0192	3.256	0.0089	0.2022	8.2206

A Pie chart was used to find out the distribution of each Losses:



Condition 2

At 102 KHz frequency.

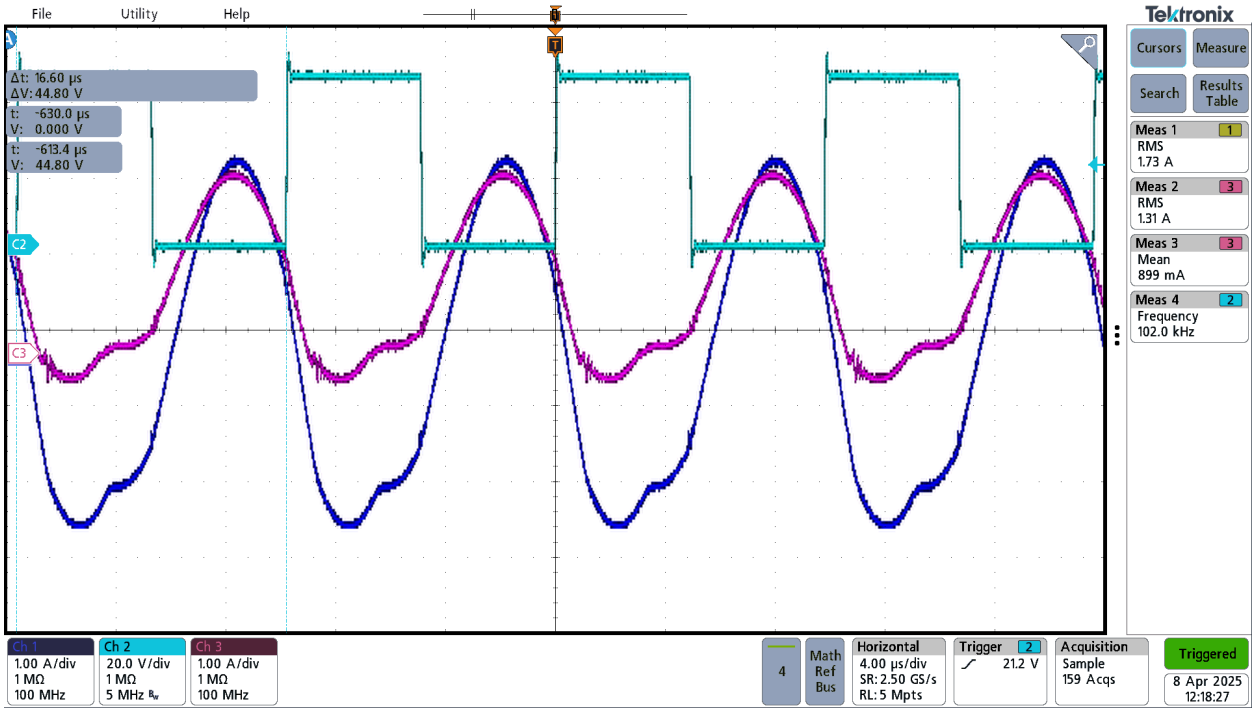


Figure: Waveforms at 102KHz

Measured RMS value of the inverter current is 1.73 A

Measured RMS value of the diode current is 1.31 A

Measured average value of the diode current is 899 mA

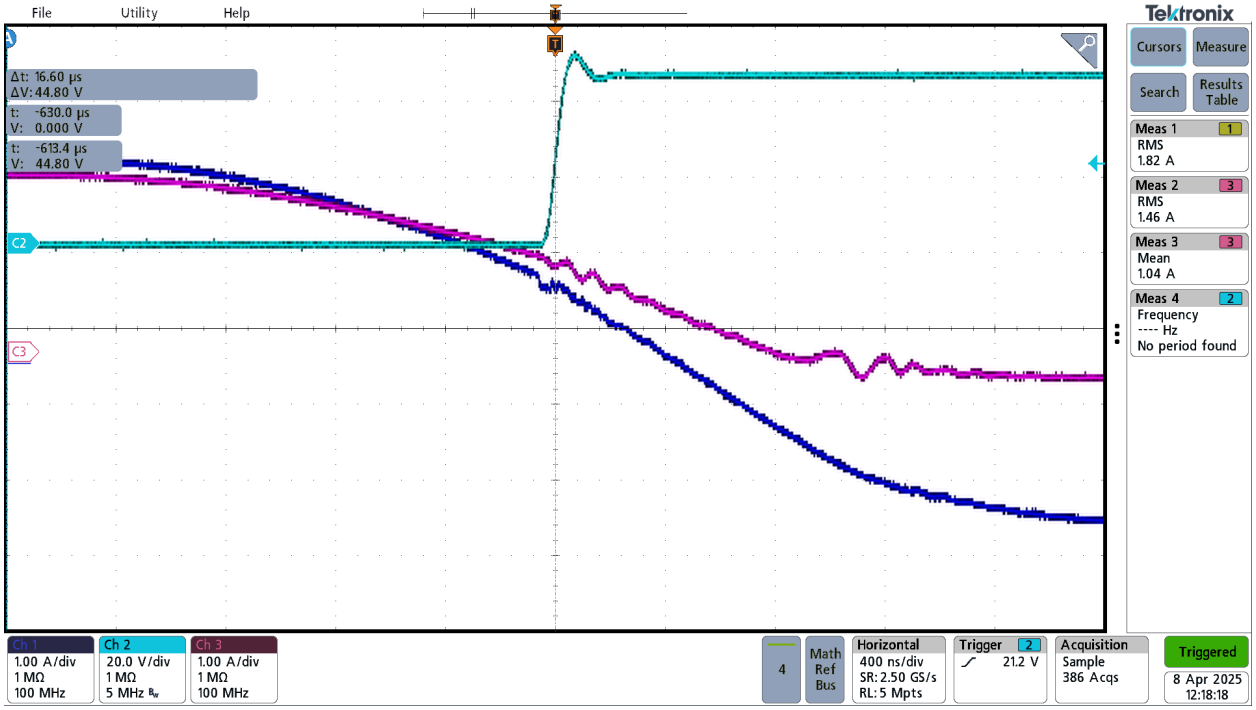


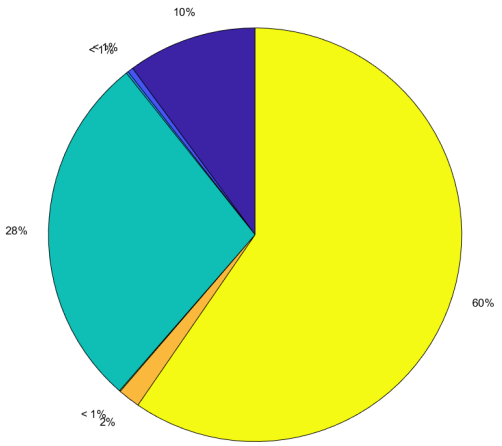
Figure: ZVS at 102 KHz

LOSS CALCULATION:
Total losses = 10.26 W

The same formula above where used for currents at 102KHz to find the power loss.

The losses for 102KHz were found out to be:

Frequency (HZ)	Diode Loss (W)	Inductor Loss (W)	Resonant capacitor Loss(W)	Filter Capacitor Loss(W)	MOSFET Conduction Loss(W)	Transformer Losses(W)	MOSFET Switching Losses(W)
102	1.0368	0.045	0.0168	2.8619	0.0078	0.1771	6.1144



Condition 3:

At 103 Khz

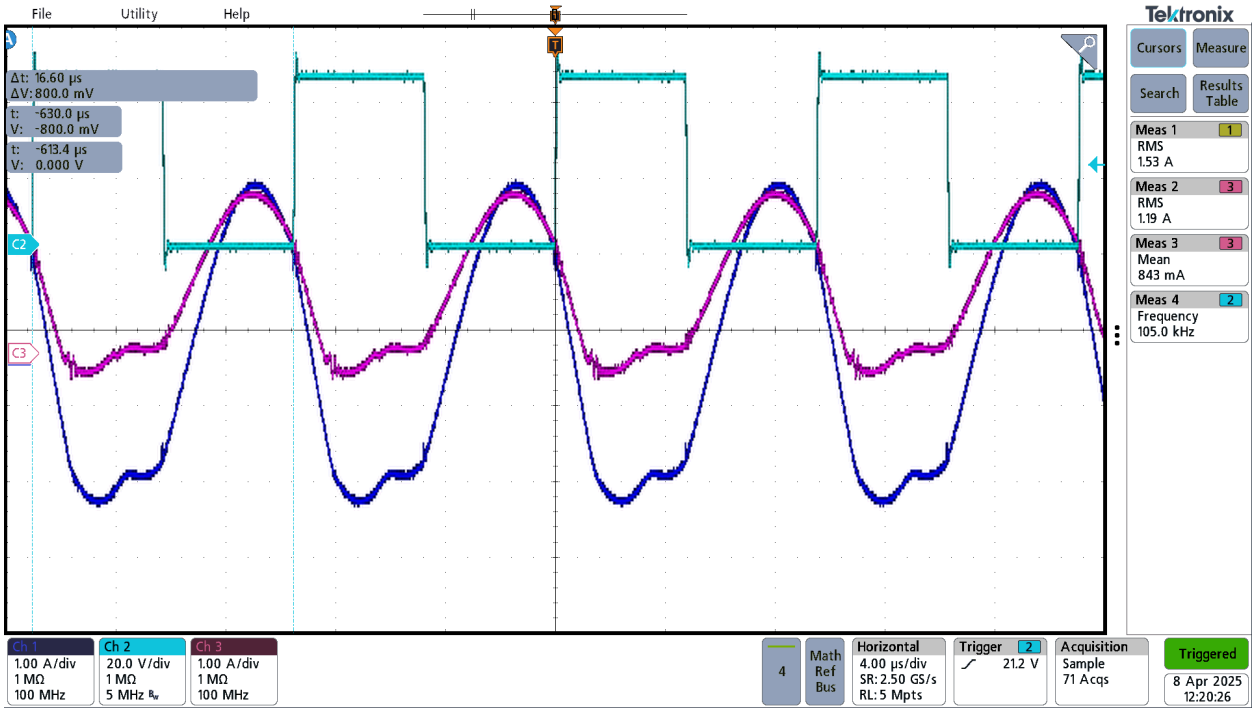


Figure: Waveforms at 105KHz

Measured RMS value of the inverter current is 1.53 A

Measured RMS value of the diode current is 1.19 A

Measured average value of the diode current is 843 mA

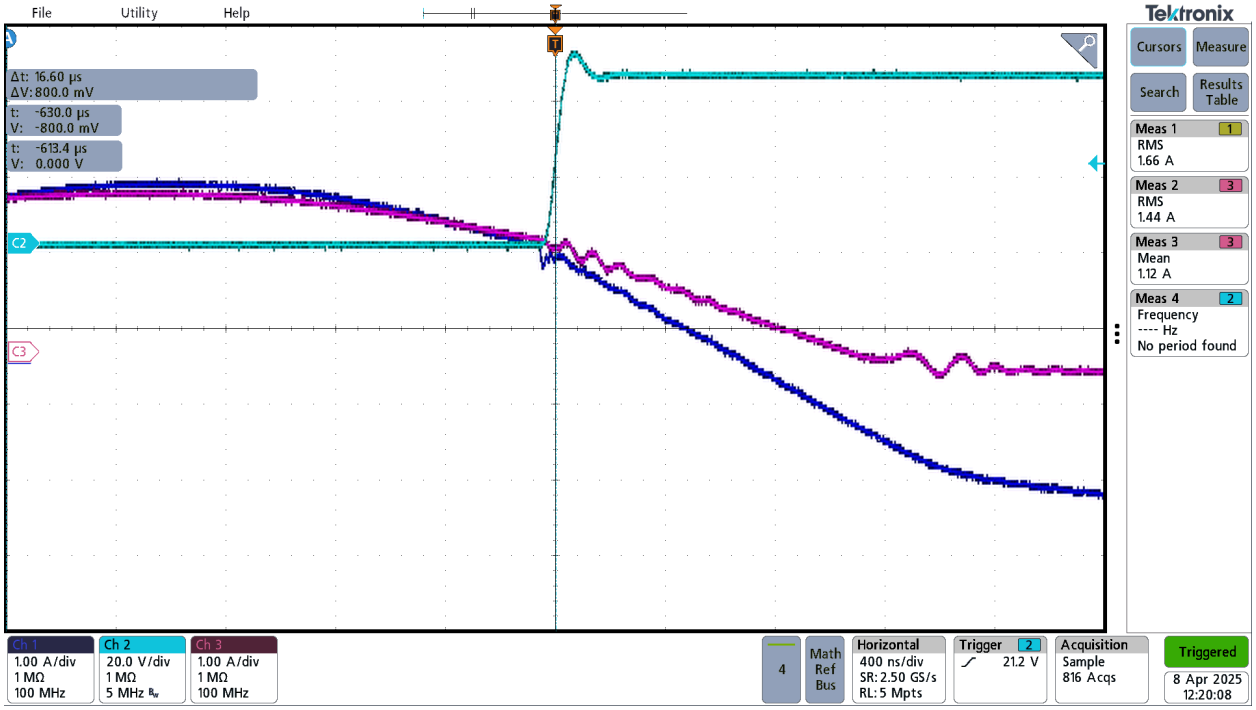


Figure: ZVS lost at 105KHz

The Loss distribution in this task cannot be estimated correctly as there were discrepancies from the original value due to connections which reduced the efficiency of the converter. But we can see that apart from switching losses <u>the Diode and the filter capacitor losses dominate</u>. The filter capacitor can be replaced with a lower ESR capacitor as the ESR for this particular converter was found out to be 3.8 ohms which is very large.